Oracle
Retrained on Homology-Restricted MegaScale
Zero-shot
Supervised, not trained on Megascale
Rectified gain @ k=96 (kcal/mol)
Predicted most stabilizing per doma
0
1
10
8.67
13.25
22.47
4.45
9.10
14.23
5.03
9.45
15.07
3.64
6.94
10.54
3.81
8.46
12.90
4.20
8.34
13.76
4.25
4.88
6.01
4.23
5.46
6.91
4.09
5.77
6.42
1.94
5.05
10.51
1.57
4.98
11.93
21.58
30.24
44.18
19.29
33.42
43.28
20.02
32.35
43.63
19.93
32.54
43.54
17.37
33.71
44.92
18.60
31.55
45.86
38.64
31.21
26.15
38.42
30.06
27.52
34.94
32.06
29.00
12.91
24.48
58.61
9.24
27.24
59.52
Top 96 per domain
ESM-MSR (p0) WT head
ESM-MSR (p0) combined
MutateEverything st1
ThermoMPNN
SPURS
ESM3-small-open unmasked
ProteinMPNN
ProSST
GeoDDG-3D
Rosetta
Δ volume ≤ −1
−1 < Δ volume ≤ 1
Δ volume > 1
Top 96 per Homology-Restricted MegaScale test domain (33 domains with the feature, 35,421 single mutants); gain = sum of max(ddG, 0) over every pick, whatever its predicted sign; boxes: interquartile range with median line; dots: domains; annotations: mean over domains; dashed lines: oracle mean.